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**H2 Mathematics (9758)**
**2026**

## 2026 H2 Math Vectors I Homework 2: Solutions with comments

True/False answers from the Basics Worksheet. Past-paper solutions: official school mark schemes where available; otherwise worked solutions (equation working). No cross-product methods used in any solution.

**1** For the following statements, state whether it is **true** or **false**.

| | **Statement** | **T/F** |
|---|---|---|
| **(a)** | $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$.<br>$a_1b_1+a_2b_2+a_3b_3=b_1a_1+b_2a_2+b_3a_3$ | **T** |
| **(b)** | $\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$.<br>$\mathbf{a}\cdot\mathbf{a}=a_1^2+a_2^2+a_3^2=\lvert\mathbf{a}\rvert^2$ | **T** |
| **(c)** | $\mathbf{i}\cdot\mathbf{j}=0$.<br>$1\cdot0+0\cdot1+0\cdot0=0$ | **T** |
| **(d)** | $\mathbf{a}\cdot\mathbf{b}$ is a vector.<br>$\begin{pmatrix}1\\0\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\\0\end{pmatrix}=1$ (a scalar) | **F** |
| **(f)** | If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$.<br>$\mathbf{0}\cdot\mathbf{b}=0b_1+0b_2+0b_3=0$ | **T** |
| **(g)** | $(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$.<br>$(2\mathbf{a})\cdot(3\mathbf{b})=2\cdot3\,(\mathbf{a}\cdot\mathbf{b})$ | **T** |
| **(h)** | $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector.<br>$(\mathbf{a}\cdot\mathbf{b})\mathbf{c}=\lambda\mathbf{c},\ \lambda=\mathbf{a}\cdot\mathbf{b}\in\mathbb{R}$ | **T** |
| **(i)** | $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$, where $\theta$ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$.<br>Definition of scalar product | **T** |

| | **Statement** | **T/F** |
|---|---|---|
| **(j)** | $\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}\cdot$<br>$\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=2+6+15=23$ (a scalar) | **F** |
| **(k)** | If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.<br>$\mathbf{i}\cdot\mathbf{j}=0,\ \mathbf{i}\neq\mathbf{0},\ \mathbf{j}\neq\mathbf{0}$ | **F** |
| **(l)** | $\lvert\mathbf{a}\cdot\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.<br>$\lvert\mathbf{i}\cdot\mathbf{j}\rvert=0\neq1=\lvert\mathbf{i}\rvert\lvert\mathbf{j}\rvert$ | **F** |
| **(m)** | $\mathbf{a}\cdot\mathbf{b}\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.<br>$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$ | **T** |
| **(n)** | For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute.<br>$\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta>0\ \therefore\ \cos\theta>0,\ 0^\circ\leq\theta<90^\circ$ | **T** |
| **(o)** | For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is obtuse.<br>$\cos\theta<0\ \therefore\ 90^\circ<\theta\leq180^\circ$ | **T** |
| **(q)** | $\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar.<br>$\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore\ \mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is meaningless | **F** |
| **(r)** | $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+\lvert\mathbf{b}\rvert^2$.<br>$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+2\mathbf{a}\cdot\mathbf{b}+\lvert\mathbf{b}\rvert^2;\ \mathbf{a}=\mathbf{b}=\mathbf{i}:\ 4\neq2$ | **F** |
| **(s)** | For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel.<br>$\cos\theta=1\ \therefore\ \theta=0^\circ$ | **T** |
| **(t)** | $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ also holds if $\theta$ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$.<br>$\cos(360^\circ-\theta)=\cos\theta$ | **T** |
| **(u)** | Let $\theta$ be the angle $ABC$. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$.<br>$\overrightarrow{BA}$, $\overrightarrow{BC}$ both start at $B$ | **T** |

| | **Statement** | **T/F** |
|---|---|---|
| **(v)** | The length of projection of **a** onto **b** is equal to the length of projection of **b** onto **a**.<br>$\mathbf{a}=\mathbf{i},\ \mathbf{b}=2\mathbf{i}:\ \lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert=1\neq 2=\lvert\mathbf{b}\cdot\hat{\mathbf{a}}\rvert$ | **F** |
| **(w)** | If $\lvert\mathbf{a}\rvert=\lvert\mathbf{b}\rvert$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$.<br>$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2=0$ | **T** |
| **(x)** | $\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$.<br>$\mathbf{a}=\mathbf{i},\ \mathbf{b}=\mathbf{j},\ \mathbf{c}=\mathbf{k}:\ \mathbf{a}\cdot\mathbf{b}=0=\mathbf{a}\cdot\mathbf{c},\ \mathbf{b}\neq\mathbf{c}$ | **F** |
| **(y)** | Let $\theta$ be the angle $ABC$. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$.<br>$\overrightarrow{AB}\cdot\overrightarrow{BC}=-\overrightarrow{BA}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$ | **F** |
| **(z)** | Let $\theta$ be the angle $CBD$. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$.<br>$\overrightarrow{BD}\cdot\overrightarrow{CB}=-\overrightarrow{BD}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$ | **F** |

**2** [EJC 2024 Promo Q5]

$\mathbf{a}$ and $\mathbf{b}$ are vectors such that $|\mathbf{a}| = \sqrt{2}$ and $|\mathbf{b}| = \sqrt{6}$. The angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{5\pi}{6}$.

**(a)** Find the exact length of projection of $\mathbf{a}$ onto $\mathbf{b}$. [2]

**(b)** By considering $(3\mathbf{a}+2\mathbf{b})\cdot(3\mathbf{a}+2\mathbf{b})$, find the exact value of $|3\mathbf{a}+2\mathbf{b}|$. [4]

| Solution | | Comments |
|---|---|---|
| **(a)** **[2]** | $\mathbf{a}\cdot\mathbf{b} = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\frac{5\pi}{6}$<br>$= \sqrt{2}\cdot\sqrt{6}\cdot\left(-\frac{\sqrt{3}}{2}\right) = -3$<br>Length of projection of $\mathbf{a}$ onto $\mathbf{b} = \frac{\lvert\mathbf{a}\cdot\mathbf{b}\rvert}{\lvert\mathbf{b}\rvert}$<br>$= \frac{\lvert-3\rvert}{\sqrt{6}} = \frac{3}{\sqrt{6}} = \frac{\sqrt{6}}{2}.$ | *Worked solution (no official student script located; working matches Vectors I Test Review).* |
| **(b)** **[4]** | $(3\mathbf{a}+2\mathbf{b})\cdot(3\mathbf{a}+2\mathbf{b}) = 9\lvert\mathbf{a}\rvert^2+12\mathbf{a}\cdot\mathbf{b}+4\lvert\mathbf{b}\rvert^2$<br>$= 9(2)+12(-3)+4(6)$<br>$= 18-36+24 = 6$<br>$\lvert3\mathbf{a}+2\mathbf{b}\rvert = \sqrt{6}.$ | |

**3** [SAJC 2024 Promo Q9(a)]

Relative to the origin $O$, the points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel. It is given that $|\mathbf{a}|=9$, $|\mathbf{b}|=1$ and $|2\mathbf{a}+\mathbf{b}|=2\sqrt{74}$.

**(i)** By considering $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$. [3]

**(ii)** Give a geometrical meaning of $|\mathbf{a}\cdot\mathbf{b}|$. [1]

**(iii)** The points $P$, $Q$ and $R$ have position vectors $7\mathbf{a}-5\mathbf{b}$, $6\mathbf{a}+5\mathbf{b}$ and $9\mathbf{a}+\lambda\mathbf{b}$ respectively. Given that these three points are collinear, find the value of $\lambda$. [3]

| Solution | | Comments |
|---|---|---|
| **(i)** **[3]** | $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})=\|2\mathbf{a}+\mathbf{b}\|^2$<br>$4\|\mathbf{a}\|^2+4\mathbf{a}\cdot\mathbf{b}+\|\mathbf{b}\|^2=(2\sqrt{74})^2=296$<br>$4(81)+4\mathbf{a}\cdot\mathbf{b}+1=296$<br>$325+4\mathbf{a}\cdot\mathbf{b}=296$<br>$4\mathbf{a}\cdot\mathbf{b}=-29$<br>$\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$ (shown). | *Worked solution (no official student script located).* |
| **(ii)** **[1]** | Since $\|\mathbf{b}\|=1$, $\|\mathbf{a}\cdot\mathbf{b}\|$ is the length of the projection of $\mathbf{a}$ onto $\mathbf{b}$. | |
| **(iii)** **[3]** | $\overrightarrow{PQ}=(6\mathbf{a}+5\mathbf{b})-(7\mathbf{a}-5\mathbf{b})=-\mathbf{a}+10\mathbf{b}$<br>$\overrightarrow{PR}=(9\mathbf{a}+\lambda\mathbf{b})-(7\mathbf{a}-5\mathbf{b})=2\mathbf{a}+(\lambda+5)\mathbf{b}$.<br>Since $P$, $Q$, $R$ are collinear and $\mathbf{a}$, $\mathbf{b}$ are non-parallel, $\overrightarrow{PR}=k\overrightarrow{PQ}$ for some scalar $k$:<br>$2\mathbf{a}+(\lambda+5)\mathbf{b}=k(-\mathbf{a}+10\mathbf{b})$<br>$2=-k,\quad \lambda+5=10k$<br>$k=-2,\quad \lambda+5=-20$<br>$\lambda=-25.$ | |

**4** [CJC 2022 Promo Q9(i),(ii)]

Relative to the origin $O$, the position vectors of points $A$, $B$ and $C$ are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where $\mathbf{b}$ is a unit vector. It is given that $\mathbf{b}$ and $\mathbf{b}-\mathbf{a}$ are perpendicular and $C$ lies on $AB$ such that $AC:CB=3:1$.

**(i)** Show that $|\mathbf{a}|=\sec\theta$, where $\theta$ is the angle between $\mathbf{a}$ and $\mathbf{b}$. [3]

By expressing $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$,

**(ii)** find the value of $|\mathbf{c}\cdot\mathbf{b}|$ and state the geometrical interpretation of $|\mathbf{c}\cdot\mathbf{b}|$. [4]

| **Solution** | | **Comments** |
|---|---|---|
| **(i)** **[3]** | $\mathbf{b}\cdot(\mathbf{b}-\mathbf{a})=0$<br>$\lvert\mathbf{b}\rvert^2-\mathbf{a}\cdot\mathbf{b}=0$<br>$\mathbf{a}\cdot\mathbf{b}=1$ (since $\mathbf{b}$ is a unit vector)<br>$\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta=1$<br>$\lvert\mathbf{a}\rvert\cos\theta=1$<br>$\lvert\mathbf{a}\rvert=\sec\theta$ (shown). | *Official solution (CJC markers' report).* |
| **(ii)** **[4]** | By the ratio theorem, $\mathbf{c}=\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}$.<br>$\mathbf{c}\cdot\mathbf{b}=\left(\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}\right)\cdot\mathbf{b}=\frac{1}{4}\mathbf{a}\cdot\mathbf{b}+\frac{3}{4}\lvert\mathbf{b}\rvert^2=\frac{1}{4}(1)+\frac{3}{4}=1$<br>$\lvert\mathbf{c}\cdot\mathbf{b}\rvert=1$.<br>$\lvert\mathbf{c}\cdot\mathbf{b}\rvert$ is the length of the projection of $\mathbf{c}$ onto $\mathbf{b}$. | |

**5** [SAJC 2025 Promo Q8(i)]

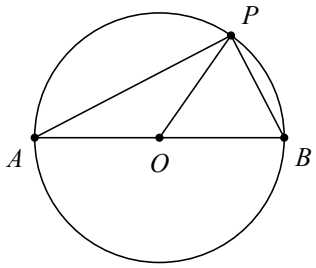
Relative to the origin $O$, two points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. It is given that $|\mathbf{a}|=2$, $|\mathbf{b}|=1$ and $|3\mathbf{a}-2\mathbf{b}|=\sqrt{37}$.

**(i)** By considering the scalar product $(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=\frac{1}{4}$, and give the geometrical meaning of $|\mathbf{a}\cdot\hat{\mathbf{b}}|$. [4]

| **Solution** | | **Comments** |
|---|---|---|
| **(i)**<br>**[4]** | $(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})=\|3\mathbf{a}-2\mathbf{b}\|^2$<br>$9\|\mathbf{a}\|^2-12\mathbf{a}\cdot\mathbf{b}+4\|\mathbf{b}\|^2=37$<br>$9(4)-12\mathbf{a}\cdot\mathbf{b}+4(1)=37$<br>$40-12\mathbf{a}\cdot\mathbf{b}=37$<br>$\mathbf{a}\cdot\mathbf{b}=\frac{1}{4}$ (shown).<br>$\|\mathbf{a}\cdot\hat{\mathbf{b}}\|$ is the length of the projection of $\mathbf{a}$ onto $\mathbf{b}$. | *Worked solution (no official student script located).* |

**6** [EJC 2025 Prelim P1 Q6]

There are three distinct points $A$, $B$ and $P$. The point $P$ lies on the circle with centre $O$ and diameter $AB$. Relative to the origin $O$, the position vectors of points $A$ and $P$ are **a** and **p** respectively.




**(a)** By expressing $\overrightarrow{AP}$ and $\overrightarrow{BP}$ in terms of **a** and **p**, show that $\overrightarrow{AP}\cdot\overrightarrow{BP}=0$. [3]

The point $C$ with position vector **c** lies on $AB$ produced such that $AC:AB$ is $\lambda:1$.

**(b)** Find **c** in terms of $\lambda$ and **a**. [2]

**(c)** It is given that $PC$ is a tangent to the circle and that angle $AOP$ is $120^\circ$. Using a suitable scalar product, find the value of $\lambda$. [4]

| Solution | | Comments |
|---|---|---|
| **(a)** **[3]** | $\overrightarrow{AP}=\mathbf{p}-\mathbf{a}, \qquad \overrightarrow{BP}=\mathbf{p}+\mathbf{a}$ (since $\overrightarrow{OB}=-\mathbf{a}$)<br>$\overrightarrow{AP}\cdot\overrightarrow{BP}=(\mathbf{p}-\mathbf{a})\cdot(\mathbf{p}+\mathbf{a})=\lvert\mathbf{p}\rvert^2-\lvert\mathbf{a}\rvert^2=0$ (since $\lvert\mathbf{a}\rvert=\lvert\mathbf{p}\rvert=$ radius). | *Official solution.* |
| **(b)** **[2]** | Since $AC:AB=\lambda:1$ and $\overrightarrow{AB}=-2\mathbf{a}$,<br>$\overrightarrow{AC}=\lambda\overrightarrow{AB}=-2\lambda\mathbf{a}$<br>$\mathbf{c}=\mathbf{a}+\overrightarrow{AC}=(1-2\lambda)\mathbf{a}$. | |
| **(c)** **[4]** | Since $PC$ is a tangent, $\overrightarrow{OP}\perp\overrightarrow{PC}$:<br>$\mathbf{p}\cdot((1-2\lambda)\mathbf{a}-\mathbf{p})=0$<br>$(1-2\lambda)\,\mathbf{p}\cdot\mathbf{a}-\lvert\mathbf{p}\rvert^2=0$<br>$(1-2\lambda)\,\lvert\mathbf{a}\rvert\lvert\mathbf{p}\rvert\cos 120^\circ-\lvert\mathbf{a}\rvert^2=0$<br>$(1-2\lambda)\,\lvert\mathbf{a}\rvert^2\left(-\frac{1}{2}\right)-\lvert\mathbf{a}\rvert^2=0$<br>$-\frac{1}{2}(1-2\lambda)=1$<br>$1-2\lambda=-2$<br>$\lambda=\frac{3}{2}$. | |

**7** [NJC 2025 Prelim P2 Q3]

With respect to an origin $O$, points $P$ and $Q$ have variable position vectors $\mathbf{p}$ and $\mathbf{q}$ respectively, given by

$$\mathbf{p} = (\cos t)\,\mathbf{i} + (\sin t)\,\mathbf{j} - \mathbf{k}, \qquad \mathbf{q} = (\cos 2t)\,\mathbf{i} - (\sin 2t)\,\mathbf{j} + \alpha\mathbf{k},$$

where $\alpha$ is a real parameter and $0 < t < \pi$.

**(i)** Find the exact values of $t$ and $\alpha$ if $O$, $P$ and $Q$ are collinear. [5]

It is given that $\alpha = 0.5$.

**(ii)** Show that $\mathbf{p} \cdot \mathbf{q} = \cos 3t - 0.5$. Find the greatest length of projection of $\mathbf{q}$ onto $\mathbf{p}$. [4]

| Solution | | Comments |
|---|---|---|
| **(i)**<br>**[5]** | Since $O$, $P$, $Q$ are collinear, $\mathbf{q} = k\mathbf{p}$ for some scalar $k$:<br>$\begin{pmatrix} \cos 2t \\ -\sin 2t \\ \alpha \end{pmatrix} = k \begin{pmatrix} \cos t \\ \sin t \\ -1 \end{pmatrix}$.<br>From the **j**-component:<br>$-\sin 2t = k \sin t$<br>$-2\cos t \sin t = k \sin t$<br>$\sin t(-2\cos t - k) = 0$<br>$\sin t = 0$ (reject, since $0 < t < \pi$)<br>or $\cos t = -\frac{k}{2}$.<br>From the **i**-component:<br>$\cos 2t = k \cos t$<br>$2\cos^2 t - 1 - k\cos t = 0$.<br>Substitute $\cos t = -\frac{k}{2}$:<br>$2\left(-\frac{k}{2}\right)^2 - k\left(-\frac{k}{2}\right) - 1 = 0$<br>$\frac{k^2}{2} + \frac{k^2}{2} - 1 = 0$<br>$k^2 = 1$<br>$k = \pm 1$.<br>From the **k**-component, $\alpha = -k$.<br>$k = 1 \Rightarrow \cos t = -\frac{1}{2} \Rightarrow t = \frac{2\pi}{3}$ $(0 < t < \pi)$, $\alpha = -1$<br>$k = -1 \Rightarrow \cos t = \frac{1}{2} \Rightarrow t = \frac{\pi}{3}$, $\alpha = 1$.<br>Hence $t = \frac{\pi}{3}$, $\alpha = 1$ or $t = \frac{2\pi}{3}$, $\alpha = -1$. | *Official solution (Method 2: scalar multiple; no cross product used).* |

| | | |
|---|---|---|
| **(ii)**<br>**[4]** | $\mathbf{p}\cdot\mathbf{q} = \cos t\cos 2t - \sin t\sin 2t - 0.5$<br>$= \cos(2t+t) - 0.5$<br>$= \cos 3t - 0.5$ (shown).<br><br>$\lvert\mathbf{p}\rvert = \sqrt{\cos^2 t + \sin^2 t + 1} = \sqrt{2}$<br>Length of projection of $\mathbf{q}$ onto $\mathbf{p} = \dfrac{\lvert\mathbf{p}\cdot\mathbf{q}\rvert}{\lvert\mathbf{p}\rvert}$<br>$= \dfrac{\lvert\cos 3t - 0.5\rvert}{\sqrt{2}}$.<br>The greatest value of $\lvert\cos 3t - 0.5\rvert$ on $0 < t < \pi$ is 1.5 (when $\cos 3t = -1$), so<br>greatest length $= \dfrac{1.5}{\sqrt{2}} = \dfrac{3}{2\sqrt{2}}$ $(= 1.06$ (3 s.f.)). | |

**END OF SOLUTIONS**